⇑Sagar Panchal
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## About

Engineer with 10+ years experience across Automation, Machine-Software, Robotics, Manufacturing, and Embedded Systems, delivering production grade solutions in complex industrial environments. I bring system level thinking and take ownership of designing, building, and architecting impactful engineering solutions.

## Education

MS, Computer engineering (3.41/4) – Dec-17
New York University NYU
BE, Instrumentation & Controls Engineering (7.04/10) – Jun-14
Gujarat Technological University

## Skills

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|---|---|
| Automation: | Machine Automation - PC based Systems, |
| | Industrial Automation - PLC/HMI based Systems, |
| | Embedded, Distributed & Real Time Systems (VxWorks) |
| | Computer Vision (Cognex, Keyence, OpenCV) & Image Processing |
| Software: | C, CPP, CS, Java, Ladder, FBD, Web - HTML, CSS, JS |
| | SCM : Git (Bash & Sourcetree) |
| | AI driven workflows - Python, Swift |

## Professional Experience

### Universal Instruments, NY

#### Software Engineer III - Automation — *Sep 22 - present*

Core Responsibilities:

- Maintain and enhance high-precision, high-speed software modules for Fuzion SMT Platform.
- Provide remote production-line support by investigating issues, analyzing logs, replicating site conditions to perform root-cause analysis, reviewing configurations and delivering software fixes, patches, and release updates.
- Enable field and application engineers to correctly set up and validate applications by offering technical guidance, interpreting machine data and supplying required configuration or software updates.

Key contributions:

- Built the SBS feature integrating Keyence SR-X300, delivering Fuzion's first side-barcode scanning capability with live-view, traceability, and error-recovery support.
- Upgraded the AMS workflow from one point to 13 locations on wafer, delivering a major accuracy improvement in wafer placement and calibration.
- Built the wafer-to-board mapping system that converts customer wafer maps into Fuzion board templates, enabling direct use of wafer data and eliminating hours of manual template creation.
- Delivered Selective Placement, enabling Fuzion's first smart pick/buffer/place workflow using DFF-based categories and avoiding-priority rules, improving overall product quality.
- Implemented AVP, adding user-defined fiducial capability beyond fixed shapes and improving alignment flexibility across ECS, NPI, and Production workflows.
- Integrated Cognex QuickBuild (VPP) with MCOS↔QBIT TCP/IP triggers and full reject/retry handling, enabling Dispense Quality Inspection on the Fuzion Precision-Bonder platform for the first time.
- Delivered key platform enhancements — Light Intensity Controller, nested-tray support, isosceles fiducial, and configurable park-position — among many other features across Fuzion.
- Tech-stack : · CS · C · CPP · Git · VxWorks · Sourcetree.

### Nor-Cal controls, CA

#### ☐ Automation Engineer — *Jan 22 - Jul 22*

- Developed automation software for utility-scale solar power plants.
- Assisted commissioning and troubleshooting of PLC/SCADA systems.
- Tech stack: Ignition SCADA · GE PLC · SEL-RTAC · Red Lion

### Mini-Circuits, NY

#### ☐ Software Engineer - Machine Automation — *Feb 18 - Nov 21*

- Developed and maintained automation software for rotary machines with computer-vision inspection.
- Implemented reverse-motion support on rotary machines with a pressing arm, enabling EVT testing.
- Delivered the new shopping-cart system for online store, improving performance and customer experience.
- Built enterprise software used across internal teams to streamline workflows reducing manual efforts.
- Tech stack: Java · JavaFX · HTML · CSS · JavaScript.

### Prima Automation, Ahmedabad, INDIA

#### ☐ Design Engineer — *Jun 15 - Jan 16*

- Programmed Siemens PLCs in TIA Portal and developed HMI interfaces for industrial automation.
- Designed electrical and control panels using AutoCAD.
- Created detailed electrical schematics and wiring diagrams using E-plan.

#### ☐ Quality Control Engineer — *Jun 14 - Jun 15*

- Electrical power testing, continuity checks and full Control Panel functional verification.
- BOM verification, component & ferrule naming checked against engineering specs.
- Inspected wiring for looseness, verified terminations, and identified any physical or assembly damage.
- PLC/HMI/SCADA functional testing to confirm full panel operation before dispatch.

## Academic Work Experience

### RevMax Fleet Optimization, NY

**Developer Intern** *Sep - Dec 17*

- Engineered Doppler-radar firmware for the SidewalkIQ pedestrian-monitoring sensor.
- Supported deployment and tested sensors across multiple NYC to validate field performance.
- Improved data consistency and optimized backend storage/transfer pipelines.
- Tech stack : · Arduino-C ·

### New York university, NY

**Instructor (ITEST Program), robotics & mechatronics dpt.** *Sep - Dec 17*

- Mentored Robotics Club programmers for FIRST Robotics and VEX competitions.
- Taught foundational programming concepts to student robotics teams.

**Lead Instructor (CREST Program), Center for K12 STEM Education** *May - Aug 17*

- Mentored high-school interns in Arduino, IoT, robotics, and 3D printing & led STEM projects including:
  -Autonomous Car (Mouse in a Maze) - STM32 Board: https://github.com/sap586/RTES_Project
  -Wireless Robot Controller - Arduino: https://github.com/sap586/WirelessRobotDay5Activity

### Neoplast (I) Pvt. Ltd., Ahmedabad, INDIA

**Engineering Trainee** *Aug 13 - May 14*

- Created a dummy working-principal based demo for 'PVC Compounding & Material handling system' using Siemens S700 and TIA-Portal.

This document is prepared using LaTeX. Click here to download the source code.